

Stanford University
Political Science 350D
Problem Set 2
Due April 13, 2015

1. Suppose $y \sim \text{Normal}(0, \sigma^2)$.
 - (a) What is the information $\mathcal{I}(\sigma^2)$ in y ?
 - (b) What is Jeffreys' prior for σ^2 ?
 - (c) Suppose you observe $y = (-2, 3, 0, -1, 2)$. What is the posterior distribution of σ^2 using the prior from part (b)?
 - (d) Compute the posterior mean and mode of σ^2 using your answer from part (c).
2. Exercise 2.11.8 in *BDA3*.
3. Exercise 2.11.13 in *BDA3*.